

SUMMARY

I am applied economist working in development, agricultural, and environmental issues, with main research focused on total factor productivity dynamics, impacts of rural infrastructure, and climate–development interactions. My work applies quasi-experimental methods, spatial and microeconomic analysis to generate evidence relevant to policy and development practice. I have worked across academic and international policy environments, contributing to research design, field implementation, analytical work, and capacity-building activities in multiple countries. My experience includes collaboration with researchers, policymakers, and practitioners, as well as engagement in both technical and operational settings through field missions, training workshops, and interdisciplinary research initiatives.

AREAS OF RESEARCH INTEREST

Primary areas: Development Economics; Agricultural and Environmental Economics; Health Economics

Secondary areas: Impact Evaluation, Applied Spatial and Microeconometrics

EDUCATION

PhD, Economics , Swedish University of Agricultural Sciences, Uppsala-Sweden	2014-2019
Visiting PhD student , University of Western Australia, Perth-Australia	June-Aug. 2018
MSc., Economics , Swedish University of Agricultural Sciences, Uppsala-Sweden	2010-2013
MSc., Rural Development , Swedish University of Agricultural Sciences, Uppsala-Sweden	2009-2011
BA., Economics , Jimma University, Jimma, Ethiopia	2004-2007

EMPLOYMENT HISTORY

Economist - United Nations Food and Agriculture Organization, Rome, Italy	Jan. 2021 to present
Postdoctoral Researcher - Swedish University of Agricultural Sciences Department of Economics, Uppsala, Sweden	Jan. 2019 - Dec. 2020
Doctoral Researcher - Swedish University of Agricultural Sciences Department of Economics, Uppsala, Sweden	Aug. 2014 - Dec. 2018
Junior Planner - Ethio Telecom, Transformation Program Office, Addis Ababa, Ethiopia	Sept. 2007 - Oct. 2009

SELECTED PEER REVIEWED ARTICLES

- (1) **Tirkaso, W.T.** (2026). On the Impact of Rural Electrification: The Role of Complementary Factors. *Journal of African Economics* (in press).
- (2) Acosta, A., **Tirkaso, W.T.**, Atravia, M., Nicolli, F., Chatzopoulos, T., Araujo-Enciso, S.R., Cicowiez, M., Eskander, S., Laxminarayan, R., and Song, J. (2026). Antimicrobial Use in Livestock: The Economic Cost of Action or Inaction. *Journal of Agricultural Economics*, 0, 1–23.
- (3) Acosta, A., **Tirkaso, W.T.**, Nicolli, F., Van Boeckel, T., and Song, J. (2025). The Future of Antimicrobial Use in Livestock. *Nature Communications*, 16(1), 2469.
- (4) **Tirkaso, W.T.**, and Hailu, A. (2022). Does Neighborhood Matter? The Effect of Spatial Proximity on Farmers Technical Efficiency Level. *Agricultural Economics*, 1–13.
- (5) Marbuah, G., Gren, I-M. and **Tirkaso, W.T.** (2021). Social capital, economic development and carbon emissions: Empirical evidence from counties in Sweden. *Renewable and Sustainable Energy Reviews*, 152, 111691.

- (6) Gren, I-M. and **Tirkaso, W.T.** (2021). Costs and equity of uncertain greenhouse gas reductions: Fuel, food and negative emissions in Sweden. *Energy Economics*, 105638.
- (7) **Tirkaso, W.T.**, and Gren, I-M. (2021). Evaluation of cost efficiency in hydropower-related biodiversity restoration projects in Sweden – a stochastic frontier approach. *Journal of Environmental Planning and Management*, 1–21.
- (8) **Tirkaso, W.T.**, and Gren, I-M. (2020). Road fuel demand and regional effects of carbon taxes in Sweden. *Energy Policy*, 144, 111648.

WORKING PAPERS AND BOOK CHAPTERS

- (1) **Tirkaso, W.T.** (2023). On the Impact of Rural Electrification: The Role of Complementary Factors. Available at SSRN: <https://ssrn.com/abstract=4377190>
- (2) **Tirkaso, W.T.**, Sandin, Leonard., Segersten, Joel, Spjut, David., and Degerman, Erik.(2018). Selected Studies in Biodiversity. In Şen, Bülent(Ed.), "Biodiversity Restoration and Renewable Energy from Hydropower: Conflict or Synergy?". pp. 120-131. ISBN:978-1-78923-233-2.
- (3) Marbuah, G., **Tirkaso, W.T.**, and Gren, I-M. (2016). Cost-effective Policy and Incomplete Information on Policy Implementation. In H., Liljenström and U, Svedin (Ed.), "Towards a Fossil-Free Society - in the Stockholm-Mälars Region". pp. 134-152. ISBN: 978-91-976048-4-0

TECHNICAL AND PROGRAMING SKILLS

- Production of Technical and Scientific Documentation: Microsoft Office and LaTeX
- Econometrics and Data Analysis Softwares: Stata, R, ArcGIS/QGIS and Python.

TEACHING EXPERIENCE

Swedish University of Agricultural Sciences, Department of Economics

Microeconomics (Bachelor level course)	Spring 2016
Environmental Economics (Bachelor level course)	Spring 2016

The Swedish Program - Stockholm School of Economics (SSE)

Energy and Environmental Economics (Bachelor level course)	Fall 2021, Spring 2022
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WORKSHOPS, CONFERENCES AND PRESENTATIONS

- (1) Technical Workshop on Antimicrobial Use Mitigation and Farmers Field School Programme Evaluation. Food and Agriculture Organization of the United Nations (FAO), Lusaka, Zambia, 16–24 August 2025.
- (2) Capacity Building Training on Productivity and Efficiency Analysis in Livestock Production. General Statistics Office of Vietnam (GSO) and the Institute of Policy and Strategy for Agriculture and Rural Development (IPSARD), Hanoi, Vietnam, 3–7 July 2023.
- (3) First Meeting of the Quadripartite Technical Group on the Economics of Antimicrobial Resistance. World Health Organization (WHO), Geneva, Switzerland, 4–5 April 2023.
- (4) The 2019 AAEA Annual Meeting. Agricultural & Applied Economics Association, Atlanta, 21–23 July 2019.
- (5) 26th Ulvön Conference on Environmental Economics. Umeå University and the Swedish University of Agricultural Sciences (SLU), Sweden, 18–20 June 2019.
- (6) EAAE Seminar 168: Behavioural Perspectives in Agricultural Economics and Management. The Swedish University of Agricultural Sciences (SLU), Sweden, 6–7 February 2019.
- (7) 2nd Nordic Annual Environmental and Resource Economics (NAERE) Workshop. Norwegian School of Economics (NHH), Norway, 12–13 April 2018.
- (8) 15th European Workshop on Efficiency and Productivity Analysis (EWEPA). Loughborough University, UK, 12–15 June 2017.

- (9) Evaluation of Cost Efficiency in Hydro-Power Related Biodiversity Restoration Projects in Sweden. Centre for Environmental and Resource Economics (CERE), Umeå University, Sweden, 25 October 2017.
- (10) International Society of Ecological Economics (ISEE) 2016 Conference. University of the District of Columbia, USA, 26–29 June 2016.
- (11) Euroleague for Life Sciences (ELLS) Scientific Student Conference. University of Hohenheim, Germany, 11–12 November 2016.
- (12) Euroleague for Life Sciences (ELLS) Scientific Student Conference. Boku University of Natural Resources and Life Sciences, Austria, 14–16 November 2013.

REFERENCES

Reference will be provided up on request.